

Xinwei Qiang

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Research interests: GPU systems, compilers, and distributed systems.

Publications

[ASPLOS '27] Impeller: Scalable and Efficient Zero-Knowledge Proof Computation Framework via Graph Compilation.

Liukun Yu*, **Xinwei Qiang***, Shixuan Sun, Chen Chen, Jingwen Leng, Jin Dong

[NeurIPS '26] AsymHP: Load-Balanced Sparse Attention for Video Diffusion Transformers.

Xinwei Qiang*, Yue Guan*, Ruihan Zhu, Mihir Jagtap, Zaifeng Pan, Zhongkai Yu, Chang Chen, Zhengding Hu, Yufei Ding, Adnan Aziz

[arXiv '26] Beyond Parallel Blindness: Information Floors and Model Gaps in Block Drafting.

Xinwei Qiang, Xiang Fang, Chang Chen, Zaifeng Pan, Yue Guan, Yufei Ding

[SC '26] FluxZK: Scalable and Efficient Zero-Knowledge Proof Computation via GPU Acceleration.

Xinwei Qiang*, Liukun Yu*, Xiyu Wang, Zhengyi Li, Shixuan Sun, Jingwen Leng, Chen Chen, Jiaping Gui, Zhenzhe Zheng, Jin Dong, Minyi Guo

[arXiv '26] GF-DiT: Scheduling Parallelism for Diffusion Transformer Serving.

Xinwei Qiang, Yifan Hu, Shixuan Sun, Jing Yang, Han Zhao, Chen Chen, Yu Feng, Jingwen Leng, Minyi Guo

[ICML '26] TritonGym: A Benchmark for Agentic LLM Workflows in Triton GPU Code.

Yue Guan*, Yichen Lin*, Xu Zhao, Jianzhu Yao, **Xinwei Qiang**, Zhongkai Yu, Pramod Viswanath, Yufei Ding, Adnan Aziz

[OSDI '26] Syncopate: Efficient Multi-GPU AI Kernels via Automatic Chunk-Centric Compute-Communication Overlap.

Xinwei Qiang*, Yue Guan*, Zhengding Hu, Keren Zhou, Yufei Ding, Adnan Aziz

[ICLR '26] DASH: Deterministic Attention Scheduling for High-throughput Reproducible LLM Training.

Xinwei Qiang, Hongmin Chen, Shixuan Sun, Jingwen Leng, Xin Liu, Minyi Guo

[SOSP '25] Mercury: Unlocking Multi-GPU Operator Optimization for LLMs via Remote Memory Scheduling.

Yue Guan, **Xinwei Qiang**, Zaifeng Pan, Daniels Johnson, Yuanwei Fang, Keren Zhou, Yuke Wang, Wanlu Li, Yufei Ding, Adnan Aziz

* Equal contribution.

Research Appointments

Research Assistant, EPCC Lab, Shanghai Jiao Tong University Jul 2024 – Jun 2026
Co-advised by Profs. [Jingwen Leng](#) and [Shixuan Sun](#)

Research Intern, Picasso Lab, University of California San Diego Jul 2025 – Dec 2025
Advisor: Prof. [Yufei Ding](#)

Education

University of California San Diego — Ph.D. in Computer Science Sept 2026 – Present
Advisor: Prof. [Yufei Ding](#)

Shanghai Jiao Tong University — B.E. in Computer Science Sept 2022 – Jun 2026
John Hopcroft Class, an honors program in computer science.

Technical Skills

Programming: C/C++, CUDA, Python, Rust, Triton